

CHAPTER 30

numerical integration



The aim of this Chapter...

...is to describe how to perform multiple integration numerically in an arbitrary number of dimensions. This is a method that is very useful when valuing certain types of high-dimensional exotic options.

In this Chapter...

- how to do numerical integration in high dimensions to calculate the price of options on baskets

30.1 INTRODUCTION

Often the fair value of an option can be written down analytically as an integral. This is certainly the case for non-path-dependent European options contingent upon d lognormal underlyings, for which we have

$$V = e^{-r(T-t)} (2\pi(T-t))^{-d/2} (\text{Det}\Sigma)^{-1/2} (\sigma_1 \dots \sigma_d)^{-1} \int_0^\infty \dots \int_0^\infty \frac{\text{Payoff}(S'_1 \dots S'_d)}{S'_1 \dots S'_d} \exp\left(-\frac{1}{2}\alpha^T \Sigma^{-1} \alpha\right) dS'_1 \dots dS'_d$$

where

$$\alpha_i = \frac{1}{\sigma_i(T-t)^{1/2}} \left(\log\left(\frac{S_i}{S'_i}\right) + \left(r - D_i - \frac{\sigma_i^2}{2}\right)(T-t) \right)$$

and Σ is the correlation matrix for the d assets and $\text{Payoff}(\dots)$ is the payoff function. Sometimes the value of path-dependent contracts can also be written as a multiple integral. American options, however, can rarely be expressed so simply.

If we do have such a representation of an option's value then all we need do to value it is to estimate the value of the multiple integral. Let us see how this can be done.



Time Out...

What is this formula?

It is nothing more complicated than the present value of something. The something is just an expected value, but, of course, the risk-neutral expected value. Interestingly, it's easier to calculate the value of this quantity than to understand all the terms in it... as you'll see below.

30.2 REGULAR GRID

We can do the multiple integration by evaluating the function on a uniform grid in the d -dimensional space of assets. There would thus be $N^{1/d}$ grid points in each direction where N is the total number of points used. Supposing we use the trapezium or mid-point rule, the error in the estimation of the integral will be $O(N^{-2/d})$ and the time taken approximately $O(N)$ since there are N function evaluations. As the dimension d increases this method becomes prohibitively slow. Note that because the integrand is generally not smooth there is little point in using a higher order method than a mid-point rule unless one goes to the trouble of finding out the whereabouts of the discontinuities in the derivatives. To overcome this 'curse of dimensionality' we can use Monte Carlo integration or low-discrepancy sequences.

30.3 BASIC MONTE CARLO INTEGRATION

Suppose that we want to evaluate the integral

$$\int \dots \int f(x_1, \dots, x_d) dx_1 \dots dx_d,$$

over some volume. We can very easily estimate the value of this by Monte Carlo simulation. The idea behind this is that the integral can be rewritten as

$$\int \dots \int f(x_1, \dots, x_d) dx_1 \dots dx_d = \text{volume of region of integration} \times \text{average } f,$$

where the average of f is taken over the whole of the region of integration. To make life simple we can rescale the region of integration to make it the unit hypercube. Assuming that we have done this

$$\int_0^1 \dots \int_0^1 f(x_1, \dots, x_d) dx_1 \dots dx_d = \text{average } f$$

because the volume is one. Such a scaling will obviously be necessary in our financial problems because the range of integration is typically from zero to infinity. I will return to this point later.

We can sample the average of f by Monte Carlo sampling using uniformly distributed random numbers in the d -dimensional space. After N samples we have

$$\text{average } f \approx \frac{1}{N} \sum_{i=1}^N f(x_i) \quad (30.1)$$

where x_i is the vector of values of $x_1, \dots, x_d$ at the i th sampling. As N increases, so the approximation improves. Expression (30.1) is only an approximation. The size of the error can be measured by the standard deviation of the correct average about the sampled average, this is

$$\sqrt{\frac{1}{N} (\bar{f}^2 - \bar{f}^2)}$$

(which must be multiplied by the volume of the region), where

$$\bar{f} = \frac{1}{N} \sum_{i=1}^N f(x_i)$$

and

$$\bar{f}^2 = \frac{1}{N} \sum_{i=1}^N f^2(x_i).$$

Thus the error in the estimation of the value of an integral using a Monte Carlo simulation is $O(N^{-1/2})$ where N is the number of points used, and is independent of the number of dimensions. Again there are N function evaluations and so the computational time is $O(N)$. The accuracy is much higher than that for a uniform grid if we have five or more dimensions.

I have explained Monte Carlo integration in terms of integrating over a d -dimensional unit hypercube. In financial problems we often have integrals over the range zero to infinity.

The choice of transformation from zero-one to zero-infinity should be suggested by the problem under consideration. Let us suppose that we have d assets following correlated random walks. The risk-neutral value of these assets at a time t can be written as

$$S_i(T) = S_i(t)e^{\left(r - D_i - \frac{1}{2}\sigma_i^2\right)(T-t) + \sigma_i\phi_i\sqrt{T-t}},$$

in terms of their initial values at time t . The random variables ϕ_i are Normally distributed and correlated. We can now write the value of our European option as

$$e^{-r(T-t)} \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} \text{Payoff}(S_1(T), \dots, S_d(T)) p(\phi_1, \dots, \phi_d) d\phi_1 \dots d\phi_d,$$

where $p(\phi_1, \dots, \phi_d)$ is the probability density function for d correlated Normal variables with zero mean and unit standard deviation. I'm not going to write down p explicitly since we won't need to know its functional form *as long as we generate numbers from this distribution*. In effect, all that I have done here is to transform from lognormally distributed values of the assets to Normally distributed returns on the assets.

Now to value the option we must generate suitable Normal variables. The first step is to generate uncorrelated variables and then transform them into correlated variables. Both of these steps have been explained above; use Box–Muller and then Cholesky. The option value is then estimated by the average of the payoff over all the randomly generated numbers.

Here is a very simple code fragment for calculating the value of a European option in NDim assets using NoPts points. The interest rate is IntRate, the dividend yields are Div(i), the volatilities are Vol(i), time to expiry Expiry. The initial values of the assets are Asset(i). The Normally distributed variables are the x(i) and the S(i) are the lognormally distributed future asset values.

```

a = Exp(-IntRate * Expiry) / NoPts
suma = 0
For k = 1 To NoPts
  For i = 1 To NDim
    If test = 0 Then
      Do
        y = 2 * Rnd() - 1
        z = 2 * Rnd() - 1
        dist = y * y + z * z
      Loop Until dist < 1
      x(i) = y * Sqr(-2 * Log(dist) / dist)
      test = 1
    Else
      x(i) = z * Sqr(-2 * Log(dist) / dist)
      test = 0
    End If
  Next i
  For i = 1 To NDim
    S(i) = Asset(i) * Exp((IntRate - Div(i) - _
      0.5 * Vol(i) * Vol(i)) * Expiry + _
      Vol(i) * x(i) * Sqr(Expiry))
  Next i

```

```

Next i
term = Payoff(S(1), S(2), S(3), S(4), S(5))
suma = suma + term
Next k
Value = suma * a

```

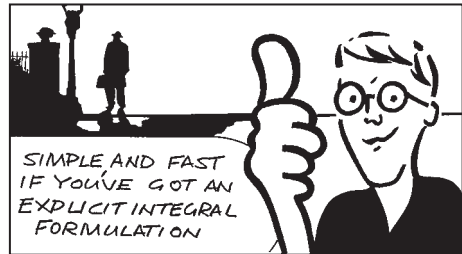
This code fragment is Monte Carlo in its most elementary form, and does not use any of the tricks described below. Some of these tricks are trivial to implement, especially those that are independent of the particular option being valued.



A more general piece of VB code

30.4 LOW-DISCREPANCY SEQUENCES

An obvious disadvantage of the basic Monte Carlo method for estimating integrals is that we cannot be certain that the generated points in the d -dimensional space are ‘nicely’ distributed. Indeed, there is inevitably a good deal of clumping. One way around this is to use a non-random series of points with better distributional properties.



Let us motivate the low-discrepancy sequence method by a Monte Carlo example. Suppose that we want to calculate the value of an integral in two dimensions and we use a Monte Carlo simulation to generate a large number of points in two dimensions at which to sample the integrand. The choice of points may look something like Figure 30.1. Notice how the points are not spread out evenly.

Now suppose we want to add a few hundred more points to improve the accuracy. Where should we put the new points? If we put the new points in the gaps between others then we increase the accuracy of the integral. If we put the points close to where there are already many points then we could make matters worse.

The above shows that we want a way of choosing points such that they are not too bunched, but nicely spread out. At the same time we want to be able to add more points later without spoiling our distribution. Clearly Monte Carlo is bad for evenness of distribution, but a uniform grid does not stay uniform if we add an arbitrary number of extra points. **Low discrepancy sequences** or **quasi-random sequences** have the properties we require.¹

There are two types of low discrepancy sequences, open and closed. The open sequences are constructed on the assumption that we may want to add more points later. The closed sequences are optimized for a given size of sample, to give the best estimate of the integral for the number of points. The regular grid is an example of a closed low-discrepancy sequence. I will describe the open sequences here.

The first application of these techniques in finance was by Barrett *et al.* (1992).² There are many such sequences with names such as **Sobol'**, **Faure**, **Haselgrove** and **Halton**. I shall describe the Halton sequence here, it is by far the easiest to describe.

¹ There is actually nothing random about quasi-random sequences.

² Andy Morton says that this has been my best piece of work, knowing full well that the numerical analysts John Barrett and Gerald Moore should have all the credit.

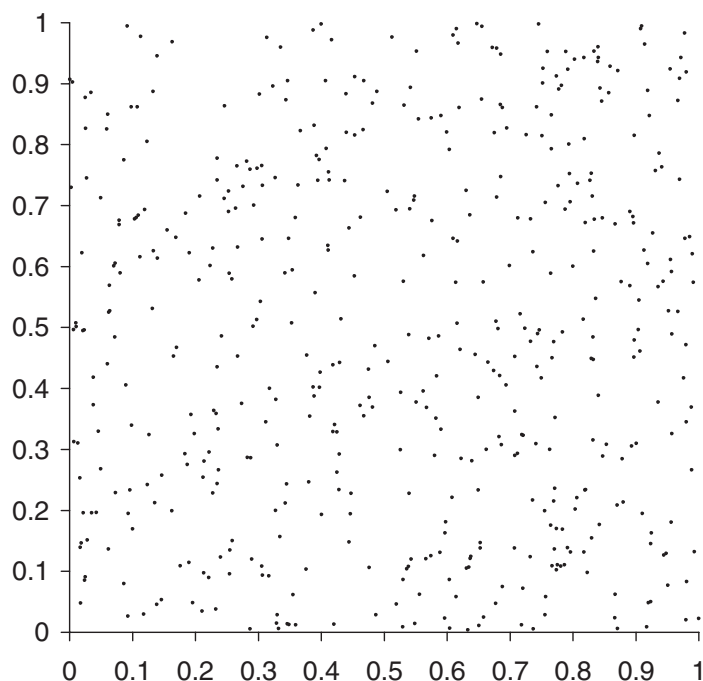


Figure 30.1 A Monte Carlo sample in two dimensions.

The Halton sequence is a sequence of numbers $h(i; b)$ for $i = 1, 2, \dots$. The integer b is the base. The numbers all lie between zero and one.³ The numbers are constructed as follows. First, choose your base. Let us choose 2. Now write the positive integers in ascending order in base 2, i.e. 1, 10, 11, 100, 101, 110, 111, etc. The Halton sequence base 2 is the reflection of the positive integers in the decimal point, i.e.

Integers base 10	Integers base 2	Halton sequence base 2	Halton number base 10
1	1	$1 \times \frac{1}{2}$	0.5
2	10	$0 \times \frac{1}{2} + 1 \times \frac{1}{4}$	0.25
3	11	$1 \times \frac{1}{2} + 1 \times \frac{1}{4}$	0.75
4	100	$0 \times \frac{1}{2} + 0 \times \frac{1}{4} + 1 \times \frac{1}{8}$	0.125
...	...	...	...

This has been called reflecting the numbers about the decimal point. If you plot the Halton points successively you will see that the next number in the sequence is always as

³ So we must map our integrand onto the unit hypercube.

far as possible from the previous point. Generally, the integer n can be written as

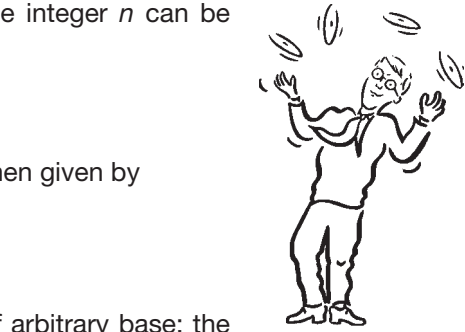
$$i = \sum_{j=1}^m a_j b^j$$

in base b , where $0 \leq a_j < b$. The Halton numbers are then given by

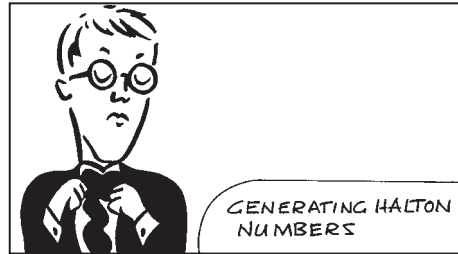
$$h(j; b) = \sum_{j=1}^m a_j b^{-j-1}.$$

Here is an algorithm for calculating Halton numbers of arbitrary base; the n th term in a Halton sequence of base b is given by $\text{Halton}(n, b)$.

```
Function Halton(n, b)
Dim n0, n1, r As Integer
Dim h As Double
Dim f As Double
n0 = n
h = 0
f = 1 / b
While (n0 > 0)
n1 = Int(n0 / b)
r = n0 - n1 * b
h = h + f * r
f = f / b
n0 = n1
Wend
Halton = h
End Function
```



Generating Halton numbers



The resulting sequence is nice because as we add more and more numbers, more and more 'dots,' we fill in the range zero to one at finer and finer levels.

In Figure 30.2 is the approximation to the Normal distribution using 500 points from a Halton sequence and the Box–Muller method. Compare this distribution with that in Figure 29.3. (You can get erroneous results if you combine the rejection-method Box–Muller with low-discrepancy numbers. See Jäckel, 2002.)

When distributing numbers in two dimensions choose, for example, Halton sequence of bases 2 and 3 so that the integrand is calculated at the points $(h(i, 2), h(i, 3))$ for $i = 1, \dots, N$. The bases in the two sequences should be prime numbers. The distribution of these points is shown in Figure 30.3, compare the distribution with that in the Figure 30.1.

The estimate of the d -dimensional integral

$$\int_0^1 \cdots \int_0^1 f(x_1, \dots, x_d) dx_1, \dots, dx_d$$

is then

$$\frac{1}{N} \sum_{i=1}^N f(h(i, b_1), \dots, h(i, b_n)),$$

where b_j are distinct prime numbers.

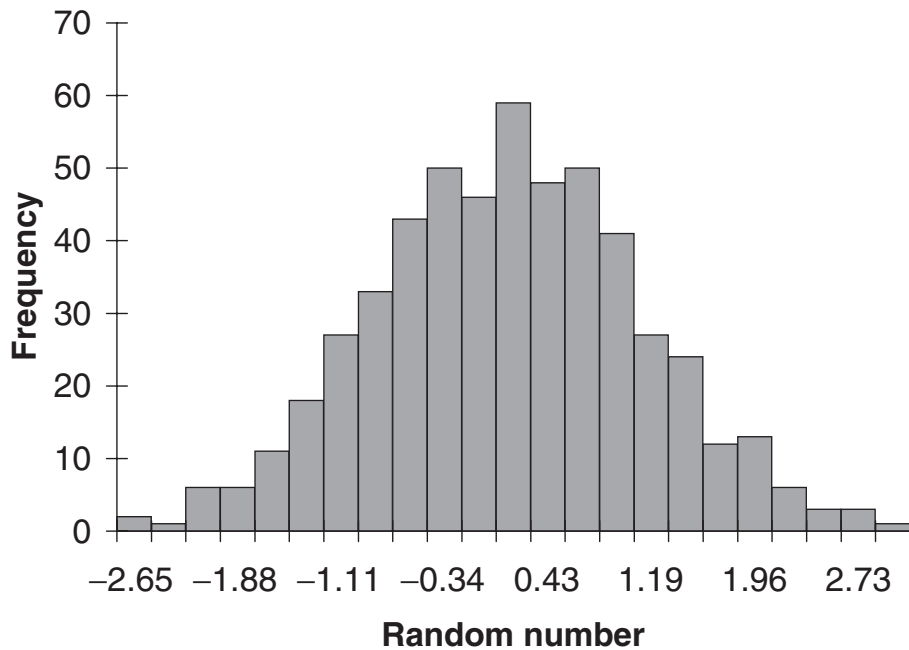


Figure 30.2 The approximation to the Normal distribution using 500 points from a Halton sequence and the Box–Muller method.

The error in these quasi-random methods is

$$O((\log N)^d N^{-1})$$

and is even better than Monte Carlo at all dimensions. The coefficient in the error depends on the particular low-discrepancy series being used. Sobol' is generally considered to be about the best sequence to use. . . but it's much harder to explain. See Press *et al.* (1992) for code to generate Sobol' points. In three or more dimensions the method beats the uniform grid.

The error is clearly sensitive to the number of dimensions d . To fully appreciate the inverse relationship to N can require an awful lot of points. However, even with fewer points, in practice the method at its worst has the same error as Monte Carlo.

In Figure 30.4 is shown the relative error in the estimate of value of a five-dimensional contract as a function of the number of points used. The inverse relationship with the Halton sequence is obvious.

Another advantage of these low-discrepancy sequences is that if you collapse the points onto a lower dimension (for example, let all of the points in a two-dimensional plot fall down onto the horizontal axis) they will not be repeated, they will not fall on top of one another. This means that if there is any particularly strong dependence on one of the variables over the others then the method will still give an accurate answer because it will distribute points nicely over lower dimensions.

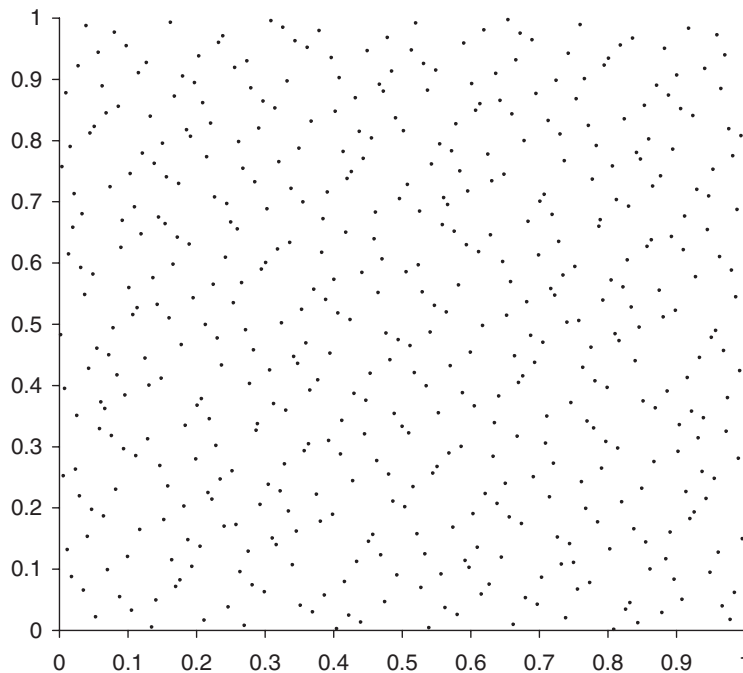


Figure 30.3 Halton points in two dimensions.

30.5 **ADVANCED TECHNIQUES**

There are several sophisticated techniques that can be used to improve convergence of Monte Carlo and related numerical integration methods. They can be generally classified as techniques for the **reduction of variance**, and hence for the increase in accuracy. None of these methods improve the speed of convergence with respect to N (for example, the error remains $O(N^{-\frac{1}{2}})$ for Monte Carlo) but they can significantly reduce the coefficient in the error term.

The method of **antithetic variables** described above is very easily applied to numerical integration. It should always be used since it can do no harm and is completely independent of the product being valued.

Control variates can also be used in exactly the same way as described above. As with the pathwise simulation for pricing, the method depends on there being a good approximation to the product having an analytic formula.

The idea behind **importance sampling** is to change variables so that the resulting integrand is as close as possible to being constant. In the extreme case, when the integrand becomes exactly constant, the ‘answer’ is simply the volume of the region in the new variables. Usually, it is not possible to do so well. But the closer one gets to having a constant integrand, then the better the accuracy of the result. The method is rarely used in finance.

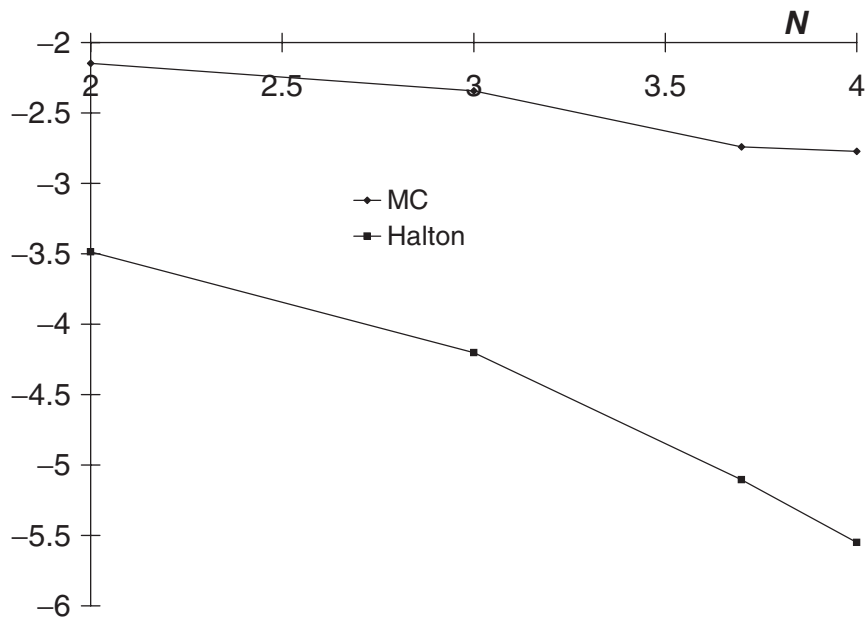


Figure 30.4 Estimate of the error in the value of a five-dimensional contract using basic Monte Carlo and a low-discrepancy sequence.

Stratified sampling involves dividing the region of integration into smaller subregions. The number of sampling points can then be allocated to the subregions in an optimal fashion, depending on the variance of the integral in each subregion. In more than one dimension it is not always obvious how to bisect the region, and can amount to laying down a grid, so defeating the purpose of Monte Carlo methods. The method can be improved upon by **recursive stratified sampling** in which a decision is made whether to bisect a region, based on the variance in the regions. Stratified sampling is rarely used in finance.

30.6 SUMMARY

Low-discrepancy sequences, where number theory meets high finance.

FURTHER READING

- See Sloan & Walsh (1990) and Stetson *et al.* (1995) for details of how to optimize a grid.
- See Barrett *et al.* (1992) for details of the Haselgrove method applied to options. And see Haselgrove (1961) for more details of the method in abstract.
- For a practical example of pricing mortgage backed securities see Ninomiya & Tezuka (1996).

- For more financial examples see Paskov & Traub (1995), Paskov (1996) and Traub & Wozniakowski (1994).
- See Niederreiter (1992) for an in-depth discussion of low-discrepancy sequences.
- See the amazing Press *et al.* (1992) for samples of code for random number generation and numerical integration. Make sure you use the latest edition, the random number generators in the first edition are not so good. They also describe more advanced integration techniques.

EXERCISES

1. Value a European call option using a Monte Carlo simulation. Use the simulation to estimate the value of an integral, as opposed to simulating the random walk for the asset.
2. Repeat the above using Halton sequences.
3. Calculate Halton sequences for bases 2, 3 and 4. Compare the results (you may wish to plot one base against another). What do you notice? How should this affect the bases you choose for a multi-factor problem?

